

FOR IMMEDIATE RELEASE

EvoChip.ai AltiCoreAI Technology Delivers 40X Faster AI Inference Than Neural Networks in Controlled Benchmark

Revolutionary technology achieves order-of-magnitude performance gains on standard CPUs, challenging fundamental assumptions about AI infrastructure requirements

Dana Point- California– February 5, 2026 – EvoChip.ai, a compute-architecture innovator redefining AI efficiency, today announced results from a controlled benchmark study demonstrating that its AltiCoreAI technology delivers 13 to 41 times faster inference performance than highly optimized neural network implementations running on standard CPU hardware.

The benchmark, conducted jointly with SidePath, a premier IT solutions provider, tested AltiCoreAI against multiple TensorFlow-based implementations across seven diverse public datasets on both workstation and server-class hardware. AltiCoreAI consistently outperformed the fastest evaluated neural network configuration in every dataset tested, sustaining 472 to 575 million inferences per second on server-grade CPUs compared to 21 to 54 million inferences per second for optimized neural networks.

"For years, the AI industry has operated on the assumption that you need massive computational resources and specialized hardware to run meaningful AI workloads," said Alain Blancquart, CEO of EvoChip.ai. "This benchmark demonstrates that assumption is wrong. We've proven that a fundamentally different mathematical approach can deliver comparable accuracy with a fraction of the computational footprint—unlocking dramatic cost savings and enabling AI deployment in environments where conventional approaches simply cannot run."

Paradigm Shift in AI Economics

The performance advantage stems from AltiCoreAI's fundamentally different architectural approach. Unlike incremental optimizations to existing neural network stacks, AltiCoreAI employs mathematically distinct inference methods

that leverage what computers do natively—fast logical operations—while minimizing dependence on computationally intensive arithmetic.

In these benchmarks the resulting efficiency gains were substantial:

- **35 to 301 times fewer parameters** than comparable neural network models
- **40 to 343 times fewer arithmetic operations** per inference
- **Order-of-magnitude throughput advantages** across all tested workloads
- **Comparable accuracy** to neural network baselines

"These aren't marginal improvements—they're structural advantages that translate directly into lower cost per decision, higher capacity per server, and broader deployment reach," added Alain Blancquart "For organizations where AI inference represents a major cost center, this magnitude of efficiency gain fundamentally changes the economics."

Real-World Deployment Implications

The benchmark's implications extend beyond raw performance metrics. AltiCoreAI's compact models and reduced input requirements enable AI deployment in resource-constrained environments where conventional neural networks cannot operate effectively including edge devices, embedded systems, and disconnected environments.

Key deployment advantages include:

- **Dramatic cost reduction:** 10-50X throughput gains translate to proportionally lower infrastructure requirements
- **Broader deployment reach:** Compact models enable AI on microcontrollers, sensors, and edge devices
- **Simplified integration:** Reduced input requirements (as few as 5-10 variables vs. 22-31 for neural networks) lower operational complexity
- **Energy efficiency:** Fewer computational operations drive proportionally lower power consumption

"The difference between requiring a \$30,000 hardware acceleration and running efficiently on a \$50 processor isn't just economic—it's about where AI can exist in the world," said Patrick O'Neill Co-Founder and Chief Technology Officer at EvoChip.ai. "We're enabling AI deployment in agricultural equipment in remote regions, medical devices in resource-limited settings, and industrial sensors in disconnected facilities. This is about making high-performance AI practical and affordable in far more places".

Robust, Comparative Benchmark

The benchmark was designed to provide credible evidence rather than optimize for isolated performance scenarios. The company with the support of SidePath, conducted testing across seven public datasets spanning credit risk assessment, fraud detection, manufacturing quality control, and medical diagnostics—reflecting the diversity of structured decisioning problems deployed in real-world systems.

AltiCoreAI was evaluated against four different neural network implementations, including C++ TensorFlow Lite with XNNPACK—a mature, highly optimized inference stack used in production environments, specifically in Edge CPU's. Each configuration underwent multiple independent tests runs under controlled conditions to ensure reproducibility and statistical validity.

“There was no cherry-pick of a narrow microbenchmark or tuning for one best case scenario” said Patrick Mulvee, CEO at SidePath. “The test was set up to be representative and repeatable, so the results are useful for real engineering/business decisions. The benchmark was structured for a fair, apples to apples comparison under consistent conditions, not a selected performance showcase”.

Target Markets and Applications

AltiCoreAI is particularly well-suited for high-volume structured decisioning workloads where predictability, reliability, and unit economics drive deployment decisions:

- **Financial Services:** Risk scoring, fraud detection, transaction monitoring, credit decisioning
- **Manufacturing:** Quality control, predictive maintenance, process optimization, defect classification
- **Healthcare:** Diagnostic support, patient risk assessment, resource allocation, clinical decision support
- **Industrial IoT:** Equipment monitoring, anomaly detection, edge analytics, autonomous systems
- **Telecommunications:** Network optimization, customer churn prediction, service quality monitoring

"We're seeing particularly strong interest from enterprises running high frequency decisioning systems where inference costs compound into millions of dollars annually," noted Jerry Conrad, Vice President of Business Development at EvoChip.ai. "For these organizations, a 20-40X efficiency gain represents

immediate, quantifiable ROI through reduced infrastructure spend and increased capacity on existing hardware."

The AltiCore Family

AltiCore is EvoChip's New mathematical framework, base of all our IP

AltiCoreAI: The umbrella brand for all AI use-case products and technologies built on AltiCore. This family include:

- **AltiCoreSWP:** Software platform optimized for laptops, desktops and servers supporting Windows, Linux, Cuda.
- **AltiCoreMCU:** A toolset that uses the model training tools of AltiCoreSWP and deploys software inference to MCU-class devices. It supports on-device training (MCU) on compatible hardware.
- **AltiCoreHDL:** FPGA/ASIC hardware inference and training.
- **AltiCoreGPU:** GPU support for accelerated training and execution (in development).
- **AltiCoreMob:** Training and inference on Android and iOS (in development)

This architectural consistency enables organizations to deploy the same core technology across cloud, edge, and embedded environments—from data center servers to battery-powered sensors—maintaining efficiency advantages throughout the deployment stack.

Importantly, AltiCoreAI supports both inference and training within the same framework, enabling devices to train or update models locally when memory and compute resources permit—supporting adaptive AI systems and reducing dependency on centralized training infrastructure.

Commercial Launch and Availability

EvoChip.ai is preparing for commercial launch in February 2026. The company is simultaneously pursuing a \$10 million equity funding to accelerate go-to-market initiatives and expand its engineering team.

"This benchmark validates our core thesis: that the AI industry has been optimizing the wrong thing," concluded Alain Blancquart "The race to build ever-larger models requiring ever-more-powerful hardware has created an AI paradigm that's simultaneously incredibly capable and incredibly inaccessible. AltiCore demonstrates that genuine artificial intelligence doesn't require

specialized hardware, massive energy consumption, or centralized infrastructure. It can run efficiently wherever you need it—and that changes everything."

Benchmark Methodology Summary

The controlled benchmark evaluated inference throughput (predictions per second) across seven public datasets on two hardware platforms:

- **Workstation CPU:** Dell Precision 8560 (Intel Core i7-13700H, 32GB RAM)
- **Server CPU:** Dell Server (Intel Xeon Gold 5416S, 64GB RAM)

Five inference methods were tested:

1. Python TensorFlow/Keras (baseline)
2. Python TensorFlow/Keras (multi-threaded, optimized)
3. C++ TensorFlow Lite with XNNPACK (highly optimized)
4. C++ TensorFlow Lite with custom parallelism
5. AltiCoreAI (multi-threaded)

Each configuration underwent multiple independent test runs under controlled conditions. Complete methodology and detailed results are available in the full benchmark report at [www.evochip.ai/benchmark].

About EvoChip.ai

EvoChip.ai is a compute-architecture company focused on redefining inference efficiency across software, edge, and hardware-integrated systems. Rather than incrementally optimizing existing neural network execution stacks, EvoChip develops mathematically distinct inference architectures designed from first principles to maximize determinism, throughput density, and energy efficiency on general-purpose and embedded compute.

Led by a CEO with over 40 years of management experience and previous \$700 million exits, EvoChip's team includes a co-founder with 10x patents and over 20 Years of algorithm and mathematics expertise, plus seasoned business development leadership with extensive financial services experience. The company's AltiCore technology portfolio—spanning AltiCoreAI (software), AltiCoreMCU (edge devices), and AltiCoreHDL (hardware)—enables efficient AI deployment from data center to sensor.

EvoChip.ai is headquartered in Dana Point, California and is preparing for commercial launch in February 2026.

For more information, visit www.evochip.ai

About SidePath

SidePath is a premier IT solutions provider and consultancy dedicated to modernizing technology infrastructures. Specializing in cloud computing, data protection, and virtualization, the company helps organizations optimize their data centers and streamline operations through strategic partnerships with industry leaders. SidePath delivers tailored, scalable solutions designed to meet the performance and security demands of today's digital landscape.

For more information, visit www.sidepath.com

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BENCHMARK RESULTS AT A GLANCE

Performance Gains on Server-Class CPU (Intel Xeon Gold 5416S)

Dataset	Speed Advantage vs. Best Neural Network
Credit Default	15.7× faster
Credit Fraud	17.2× faster
Give Me Credit	14.9× faster
Intelligent Manufacturing (High Efficiency)	18.6× faster
Intelligent Manufacturing (Low Efficiency)	19.0× faster
Machine Failure	9.1× faster
SPECT Medical Imaging	27.6× faster

Model Efficiency Comparison

Metric	AltiCoreAI Advantage
Parameters	35-301× fewer
Arithmetic Operations per Inference	40-343× fewer
Input Variables Required	50-70% fewer

Absolute Throughput Performance

Platform	AltiCoreAI	Neural Network (best)
Server CPU	472-575 million inferences/sec	21-54 million inferences/sec
Workstation CPU	225-361 million inferences/sec	17-48 million inferences/sec

Note to Editors: High-resolution images, charts, and technical diagrams are available upon request. EvoChip executives are available for interviews and technical briefings.

Forward-Looking Statements: This press release contains forward-looking statements regarding EvoChip.ai technology, commercial launch plans, and market potential. Actual results may differ materially from those projected. Performance results are based on benchmark testing under controlled conditions and may not reflect performance in all deployment scenarios.

